

Hardik Hemant Pandey

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🔗 [Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Google Cloud Profile](#)

Professional Summary

MSc Computing Science student at the University of Glasgow with hands-on experience building AI systems that prioritise privacy, efficiency, and practical utility. I engineer local LLM inference pipelines, RAG architectures, and AutoML tooling from first principles — with a focus on making on-device AI genuinely usable for real workflows. My background spans MLOps in an industrial setting (Samsung), semantic memory systems, and applied ML research. I care deeply about privacy-preserving AI: powerful models should not require constant cloud dependency.

Education

University of Glasgow, UK | MSc Computing Science | Sept 2025 – Sept 2026

Relevant Coursework: Introduction to Data Science and Systems, Programming and Systems Development, Research and Professional Skills, Human Computer Interaction Design and Evaluation, Secure Systems Programming in Rust.

Chandigarh University, India | B.E. Computer Science & Engineering | 2021 – 2025

Relevant Coursework: AI & ML, Computer Vision, Data Structures, Python, Mathematics, Probability and Statistics, DBMS, Cloud Computing and Distributed Systems, Disruptive Technologies, DSA Analysis.

CGPA: 8.14/10 | First-author IEEE publication on AI-powered scalable digital platforms.

Experience

Samsung Electro-Mechanics, Bangalore | *Student Intern* | Oct 2024 – Jun 2025

- Deployed and managed Kubernetes clusters to support multi-team machine learning workflows, improving operational efficiency.
- Developed and integrated the Python package *pado_mlops* into SEMLOPs, streamlining ML pipeline deployment and management.
- Monitored distributed system performance (Grafana, Prometheus), conducted POCs on AutoML/XAI/HPA, and collaborated with cross-functional teams to deliver milestones on schedule.
- Image repository maintenance on development and production environments.

Zooniverse (Remote, Volunteering) | *Data Annotation & Classification Contributor* | Apr 2026 – Present

- Precision Data Management: Executed 4,689 high-precision classifications across 10 distinct scientific projects, ensuring the creation of “ground truth” datasets for machine learning models.
- Systematic Analysis: Utilised pattern recognition and feature identification to support the development of high-fidelity training pipelines, ensuring data accuracy for large-scale distributed science projects.

VANKADEL (UK, Remote, Pro-bono) | *Threat Intelligence Analyst (Part-time)* | Oct 2025 – Dec 2025

- Analysed data to generate actionable insights for operational decision-making.
- Produced clear, concise reports to support strategic and operational objectives.
- Managed tasks independently, maintaining quality and documentation standards.

Projects

AI Companion: A Modular RAG-Powered Conversational Engine | Python, RAG, Gradio, Vector DB

- AI companion utilising a Retrieval-Augmented Generation (RAG) pipeline, running entirely locally with no cloud dependency.
- Dual-layer memory architecture using FAISS (vector database) for semantic long-term retrieval and JSON archiving for full conversation history.
- Built with llama-cpp-python for GGUF model inference and SentenceTransformers for high-accuracy vector embeddings.
- Dynamic persona customisation, multi-session management, and real-time knowledge extraction.
- GitHub: github.com/Hardik-7892/AI-Companion

Auto-ML Studio | Python, AutoGluon, Streamlit

- Built a no-code machine learning web app enabling non-technical users to upload datasets, trigger model training, and review evaluation results through a browser interface.
- Integrated AutoGluon to automate feature engineering, model selection, and ensembling with minimal configuration.
- GitHub: github.com/Hardik-7892/auto-ml

Password Security Enhancement Using Emojis | Python, Google Sheets API | *Project Lead & Developer*

- Designed and conducted a user study comparing text-only, emoji-only, and hybrid password systems to evaluate usability and security trade-offs.
- Built a Gradio-based data collection interface integrated with Google Sheets API for real-time participant response capture and analysis.
- Processed and analysed behavioural data to derive actionable recommendations on emoji adoption in authentication systems.
- Sole developer across research design, tool development, and data pipeline — from prototype to published findings.
- GitHub: github.com/Hardik-7892/Improving-Passwords-using-Emojis

AI-Powered Ed-Tech Platform | *Team Lead & Project Manager*

- Led a 4-member team to define product roadmap, prioritise features, and manage delivery.
- Translated user needs into system features and coordinated development milestones.
- Published IEEE research paper based on project insights.

Publications

Advance Approaches in AI-Powered Website Development for Educational Delivery via Ed-Tech

Published research examining the architectural evolution of web-based information systems and the technical challenges of integrating scalable digital platforms within complex educational ecosystems.

- DOI: [10.1109/TQCEBT59414.2024.10545211](https://doi.org/10.1109/TQCEBT59414.2024.10545211)
- Publisher: IEEE
- Role: First Author

Certifications / Awards

CoSE My Future Skills Award — University of Glasgow College of Science & Engineering | May 2026

Google Cloud ML Skills — Diamond League | 50+ skill badges including Vertex AI, BigQuery ML, and TensorFlow labs

Technical Skills

- **Product & Analytical:** Root Cause Analysis, Data Analysis, User Feedback Integration
- **Programming & Tools:** Python, C++, Rust, SQL (BigQuery), LLM Orchestration (RAG), APIs, Docker, Kubernetes, Grafana, Prometheus, Jira, Confluence, BitBucket
- **AI/ML & Data Engineering:** Vector Databases (FAISS), Embeddings (SentenceTransformers), GGUF Model Inference (llama-cpp), Semantic Search, Prompt Engineering, AutoGluon, AutoML
- **Cloud:** Google Cloud Skills Boost – Diamond League, BigQuery, GKE (Google Kubernetes Engine), Console, Compute Engine, API & Services, Pub/Sub, Looker, VPC, Logging and Monitoring

Additional

- Languages: English (Fluent), Hindi (Native)
- Interests: Chess, Reading, Music
- Right to Work in the UK
- Open to Relocate